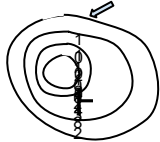
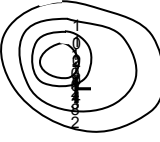
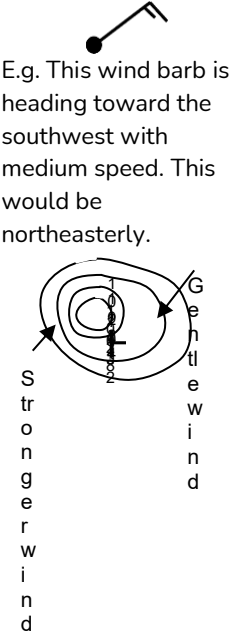


10.1 Synoptic charts

A synoptic chart is a map showing the distribution of air pressure, wind direction, and weather conditions at a given time. This is used by meteorologists to understand weather patterns and predict future weather and hazard events.

Cheat sheet

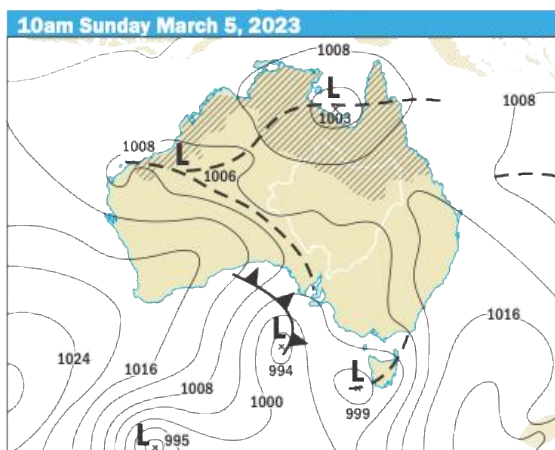
Key features	What they tell us	What they look like
Isobars	These are the lines on the map. The lines join places of equal air pressure, so everywhere on that line is the same hectopascals (hPa).	
Pressure systems: High (H) and Low (L)	Isobars also display where pressure systems are. As pressure systems move as 'cyclones', they form large circles. If the air is getting higher to the centre of the circle, it is a high pressure system. If it is getting lower, it is low!	
Winds	Using wind indicators to determine wind: Wind indicators show the strength and direction the wind comes from. Think of these like an arrow, with the dot the tip of the arrow. The more 'feathers' on the back of the arrow, the faster the wind! It points in the direction the wind is heading - but we call wind where it comes from. Using isobars to determine wind: When isobars are closer together, it indicates strong wind. When they are further apart, it means light winds. As wind follows the pressure system, they flow clockwise around lows and anticlockwise around highs (in the Southern hemisphere).	E.g. This wind barb is heading toward the southwest with medium speed. This would be northeasterly. 
Fronts	Cold and warm fronts are also important symbols on weather maps. Fronts occur when warm and cold air masses meet and interact in the atmosphere (e.g. cold ocean air meets warm continental air). Fronts indicate approaching storms or rapid weather changes. Cold fronts are symbolised with a line with little triangles on it, and warm fronts have semi-circles.	
Trough	Troughs are long, stretched-out areas of low pressure where air is rising, making the weather more cloudy, wet, and stormy. Ridges are long, stretched-out areas of high pressure where air is sinking, bringing clearer and calmer weather.	

How to interpret a weather map

1. Start by locating the isobars to identify areas of high and low pressure
2. Look at the spacing of isobars or wind indicators to determine wind strength and direction
3. Identify any fronts or troughs to predict changes in weather, such as storms or rainfall.
4. Combine all features to make an overall weather prediction. In Australia, weather systems usually move from the west or south-west towards the east over several days (often around 3–5 days). While this can change, it helps meteorologists make short-term weather predictions.

Synoptic chart - practise

Use the following synoptic chart to answer the questions.



Adapted from BOM

1. Describe the location of one high-pressure system shown on the map.

2. Describe the location of one low-pressure system shown on the map.

3. Locate one cold front and one trough of low pressure on the map. Draw an arrow to them.

4. What is the approximate air pressure above Sydney? _____

5. Describe the location of the strongest winds. _____

6. Identify one area likely to experience unstable or rainy weather. Explain why.

7. Identify one area experiencing clear and stable weather. Explain why.

8. Describe the weather conditions in Hobart, and explain your answer.

9. Describe the weather conditions in Darwin, and explain your answer.

10. Explain the direction of wind flow above Melbourne.

11. Describe how the cold front may affect weather conditions as it passes.

12. Based on the typical movement of weather systems across Australia, predict how the weather in eastern Australia is likely to change over the next 2–3 days.
